Mohamed Amine Merzouk

Postdoctoral researcher · Mila & McGill University

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Biography

AI safety researcher working on the alignment of language models. I am a postdoctoral researcher at McGill University and Mila with Pr. Adam Oberman, where I study alignment methods that act on a model's internal representations, what language models encode about their own state and outputs, and honesty and deception. I come from a cybersecurity background: my PhD thesis at Polytechnique Montréal, co-supervised by Pr. Frédéric Cuppens and Reda Yaich (IRT SystemX), addressed adversarial attacks and defenses for machine learning-based network intrusion detection and was nominated for the best thesis award.

Education

Philosophiæ doctor (PhD) in Computer Science — *Montréal, Canada*

Polytechnique Montréal — *January 2021 - September 2024*

Thesis: *A Landscape of Adversarial Threats to Machine Learning-Based Intrusion Detection*, nominated for the best thesis award. Supervisors: Pr. Frédéric Cuppens (Polytechnique Montréal) and Reda Yaich (IRT SystemX).

Master's degree in Computer Science — *Algiers, Algeria*

École Nationale Supérieure d'Informatique (ESI ex.INI) — *September 2018 - September 2020*

Thesis: *Adversarial Machine Learning against Anomaly-based Intrusion Detection Systems*. Supervisor: Pr. Lydia Bouzar.

Engineer's degree in Computer Science — *Algiers, Algeria*

École Nationale Supérieure d'Informatique (ESI ex.INI) — *September 2017 - September 2020*

Preparatory class in Mathematics and Computer Science — *Sidi Bel Abbès, Algeria*

École supérieure en informatique 08 Mai 1945 — *September 2015 - September 2017*

Professional experience

Postdoctoral researcher — *Montréal, Canada*

McGill University & Mila - Quebec Artificial Intelligence Institute — *October 2024 - Present*

AI safety and alignment of language models, with Pr. Adam Oberman.

Lecturer — *Montréal, Canada*

Polytechnique Montréal — *January 2021 - August 2026*

Undergraduate and graduate cybersecurity courses.

Affiliated Doctoral researcher — *Montréal, Canada*

IRT SystemX — *January 2021 - September 2024*

Doctoral research on adversarial machine learning for network intrusion detection.

Research intern — *Rennes, France*

IMT Atlantique — *March 2020 - September 2020*

Research internship leading to the paper *A Deeper Analysis of Adversarial Examples in Intrusion Detection* (CRiSIS 2020).

Cybersecurity audit intern — *Algiers, Algeria*

KPMG — *July 2019 - August 2019*

Implementation of the ISO 27001 information security management standard: risk assessment and drafting of security policies.

Virtualization and cloud intern — *Algiers, Algeria*

ADEX Technology — *June 2018 - November 2018*

Design and deployment of a Red Hat OpenStack private cloud: architecture of the compute and security hardening.

IT support — *Algiers, Algeria*

British Petroleum — *September 2018 - October 2018*

User support and workstation maintenance: troubleshooting, software installation and account management.

Software development intern *Algiers, Algeria*

EGC 1988 *July 2017 - October 2017*

Development of an enterprise resource planning (ERP) software.

Teaching experience

INF8102	**Security in Cloud Environments.** Lecturer: Fall 2025.
INF8085	**Cybersecurity.** Lecturer: Summer 2025, Winter 2026, Summer 2026.
CR345	**Security of Servers.** Lecturer: Winter 2025.
INF4420	**Introduction to Computer Security.** Lecturer: Summer 2023, Summer 2024. Lab instructor: Winter 2022, Fall 2022, Summer 2023, Summer 2024.
INF6103	**Cybersecurity of Critical Infrastructures.** Lab instructor: Winter 2022, Fall 2022, Fall 2023, Fall 2024.
INF8602	**Operating Systems Security.** Lab instructor: Winter 2022.
INF1040	**Introduction to Computer Engineering.** Evaluator: Winter 2021, Fall 2021.

Publications

Journal articles

Efficiency vs. Alignment: Investigating Safety and Fairness Risks in Parameter-Efficient Fine-Tuning of LLMs
Mina Taraghi, Yann Pequignot, Amin Nikanjam, Mohamed Amine Merzouk, Foutse Khomh
IEEE Transactions on Artificial Intelligence (2026). 2026

Adversarial robustness of deep reinforcement learning-based intrusion detection
Mohamed Amine Merzouk, Christopher Neal, Joséphine Delas, Reda Yaich, Nora Boulahia-Cuppens, Frédéric Cuppens
International Journal of Information Security (2024). Springer, 2024

Investigating the practicality of adversarial evasion attacks on network intrusion detection
Mohamed Amine Merzouk, Frédéric Cuppens, Nora Boulahia-Cuppens, Reda Yaich
Annals of Telecommunications 77.11 (2022) pp. 763–775. Springer, 2022

Conference proceedings

How Much is Left? LLMs Linearly Encode Their Remaining Output Length
Mohamed Amine Merzouk, Dmitri Carpov, Mirko Bronzi, Damiano Fornasiere, Adam Oberman
Advances in Neural Information Processing Systems (NeurIPS), 2026

Efficient Safety Alignment of Language Models via Latent Personality Traits
Mohamed Amine Merzouk, Nolan Smyth, Damiano Fornasiere, Linh Le, David Williams-King, Adam Oberman
Conference on Language Modeling (COLM), 2026

Latent Personality Alignment: Improving Harmlessness Without Mentioning Harms
Linh Le, David Williams-King, Mohamed Amine Merzouk, Aton Kamanda, Adam Oberman
Trustworthy AI Workshop, International Conference on Learning Representations (ICLR), 2026

Diffusion-based Adversarial Purification for Intrusion Detection
Mohamed Amine Merzouk, Erwan Beurier, Reda Yaich, Nora Boulahia-Cuppens, Frédéric Cuppens, Foutse Khomh
Proceedings of the 39th Annual IFIP WG 11.3 Conference on Data and Applications Security and Privacy (DBSec), 2025

Parameterizing poisoning attacks in federated learning-based intrusion detection
Mohamed Amine Merzouk, Frédéric Cuppens, Nora Boulahia-Cuppens, Reda Yaich
Proceedings of the 18th International Conference on Availability, Reliability and Security (ARES), 2023

Evading Deep Reinforcement Learning-based Network Intrusion Detection with Adversarial Attacks
Mohamed Amine Merzouk, Joséphine Delas, Christopher Neal, Frédéric Cuppens, Nora Boulahia-Cuppens, Reda Yaich
Proceedings of the 17th International Conference on Availability, Reliability and Security (ARES), 2022

A Deeper Analysis of Adversarial Examples in Intrusion Detection
Mohamed Amine Merzouk, Frédéric Cuppens, Nora Boulahia-Cuppens, Reda Yaich
Proceedings of the 15th International Conference on Risks and Security of Internet and Systems (CRiSIS), 2021

Preprints

GrayShield: Bit-Level Sanitization for Transformer Model Supply-Chain Security
Armstrong Foundjem, Tsung-Hsien Chuang, Foutse Khomh, Mohamed Amine Merzouk
Preprint, arXiv:2610.04319, 2026

Quantifying Overclaiming Propensity in Frontier LLM Agents
Nolan Smyth, Yorguin-Jose Mantilla-Ramos, Pascal Jr Tikeng Notsawo, Saskia Helbling, Alberto Tosato, Mohamed Amine Merzouk, Nouha Dziri, Gauthier Gidel, Tommaso Tosato
Preprint, arXiv:2609.20812, 2026

Deep-Cover Agents: Long-Horizon Prompt Injections on Production LLM Systems
Neel Alex, Mohamed Amine Merzouk, David Krueger
Preprint, 2026

Theses

A Landscape of Adversarial Threats to Machine Learning-Based Intrusion Detection
Mohamed Amine Merzouk
Ph.D. Thesis. Polytechnique Montréal, 2024

Adversarial Machine Learning against Anomaly-based Intrusion Detection Systems
Mohamed Amine Merzouk
M.Sc. Thesis. École Nationale Supérieure d'Informatique (ESI ex. INI), 2020

Awards and Funding

2025	**AI Safety Catalyst Grant (with Pr. Khomh, Pr. Ma, and Pr. Goebel),** Canadian AI Safety Institute & CIFAR	*Montréal, Canada*
2024	**Nomination for the Best Thesis Award,** Polytechnique Montréal	*Montréal, Canada*
2023	**Best Presentation Award,** InCyber International Forum	*Montréal, Canada*
2021	**Mitacs Accelerate International Graduate Research Fellowship,** Polytechnique Montréal	*Montréal, Canada*
2014	**1st Place in Regional Mathematical Olympiad,** ASA Science	*Algiers, Algeria*

Invited Talks

2025	**IRT SystemX,** Diffusion-based Adversarial Purification for Intrusion Detection	*Paris, France*
2023	**InCyber International Forum,** Trustworthy AI (Best presentation award)	*Montréal, Canada*
2023	**BSides Algiers,** Adversarial threats at training and inference time	*Algiers, Algeria*
2023	**Cyberconference (Cybereco),** Round table on AI security	*Montréal, Canada*
2022	**School of AI Summit,** Adversarial Poisoning in Federated Learning	*Algiers, Algeria*
2022	**International Cybersecurity Forum,** Adversarial Machine Learning Masterclass	*Montréal, Canada*
2022	**Canadian AI Federated Learning Workshop,** The Threat of Malicious Clients in Federated Learning	*Toronto, Canada*
2022	**Mila - Quebec Artificial Intelligence Institute,** Generating adversarial examples with constraints	*Montréal, Canada*
2021	**CRITiCAL chair launch,** Adversarial attacks against intrusion detection systems	*Paris, France*
2021	**NeurIPS Meetup,** Adversarial attacks against intrusion detection systems	*Algiers, Algeria*
2018	**OpenStack User Group Meeting,** OpenStack Security	*Algiers, Algeria*
2017	**Docker Chapter,** Security Enhanced Linux	*Algiers, Algeria*

Peer Review Activities

IEEE	Transactions on Intelligent Vehicles
Elsevier	Pattern Recognition
Elsevier	Computer Networks
Springer	International Journal of Information Security
ESORICS	European Symposium on Research in Computer Security
ISC	International Conference on Information Security
FPS	International Symposium on Foundations & Practice of Security
WPES	Workshop on Privacy in the Electronic Society
SECRYPT	International Conference on Security and Cryptography
CSNet	Cyber Security in Networking Conference
CASCON	IEEE International Conference on Collaborative Advances in Software and Computing

Mentoring

Ph.D. students	Mina Taraghi (graduated in 2026) and Masoud Karami, Polytechnique Montréal.
M.Sc. students	Maxime Gourceyraud (graduated in 2024) and Joséphine Delas (graduated in 2023), Polytechnique Montréal.
Undergraduates	Research interns Miran Qarachatani (Lakehead University), Adjobi Vernet Emmanuel Placide (Université de Moncton), Junyu Huo (University of Alberta) and Alfie Chuang (National Taiwan University).

Community and social contributions

2026-...	**Mentor,** SPAR (Supervised Program for Alignment Research)	*Online*
2026	**Program committee,** Tool Demonstration Track, 17th International Conference on Internetware (Internetware 2026)	*Gold Coast, Australia*
2026	**Program committee,** CPRA workshop in conjunction with ARES 2026	*Linköping, Sweden*
2025-...	**Organizer,** Mila's AI Safety Reading Group	*Montréal, Canada*
2025	**Co-chair,** SEMLA Symposium 2025	*Montréal, Canada*
2025	**Program committee,** CPRA workshop in conjunction with ARES 2025	*Ghent, Belgium*
2024-...	**Member,** Mila's Information Security Committee	*Montréal, Canada*
2024-...	**Founder and organizer,** SEMLA Reading Group	*Montréal, Canada*
2018-19	**President,** Shellmates (first Algerian cybersecurity club)	*Algiers, Algeria*
2021-...	**Union representative,** Association pour une Solidarité Syndicale à l'École Polytechnique (ASSEP)	*Montréal, Canada*
2023-...	**Animal rights activist,** WeTheFree	*Montréal, Canada*
2020-21	**Volunteer,** Secours Populaire Français	*Rennes, France*
2021-24	**Artistic mediator,** La Gang à Rambrou (community center for intellectual disability and autism)	*Montréal, Canada*
2021-...	**Theater actor,** Les Pains Perdus	*Montréal, Canada*
2024-...	**Choir singer,** Canopées	*Montréal, Canada*
2023-...	**Guitarist,** The Tiny Tax Band	*Montréal, Canada*

References

Adam Oberman	Postdoctoral advisor, Full Professor at McGill University and Academic Member at Mila
Frédéric Cuppens	PhD supervisor, Full Professor at Polytechnique Montréal
Reda Yaich	PhD co-supervisor, Head of the cybersecurity team at IRT SystemX
Foutse Khomh	Full Professor at Polytechnique Montréal and Academic Member at Mila
Yacine Challal	Graduate Program Director at ESI ex.INI
Fetheddine Guenaoui	Senior Manager at KPMG

Contact details are available on request.